

Max Kelsen: Using TensorFlow Research Cloud to help Australians living with cancer

Tell us your challenge. We're here to help.

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About Max Kelsen

Max Kelsen is an analytics and software engineering agency specializing in machine learning and artificial intelligence.

Industries: Healthcare, Life Sciences

Location: Australia

Using TensorFlow
Cloud, Max Kelsen
help improve the
rate of people with
by predicting their
to treatment

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TensorFlow Research Cloud

(<https://www.tensorflow.org/tfrc/>)

Google Cloud Results

- Uses AI and ML to predict how patients will respond to treatment
- Explores new cancer techniques and treatments at scale
- Provides opportunities to address other difficult healthcare issues

Aims to help improve the survival rate of people with cancer

Cancer is one of Australia's greatest health challenges. One in four individuals are diagnosed with cancer during their lives and an estimated 48,000 Australians were expected to die from the disease in 2018 alone¹.

When a patient is correctly diagnosed with cancer, the best treatments achieve a 48 percent success rate over three years². This means millions of dollars are spent on unsuccessful or partially successful cancer treatments.

Max Kelsen and Google Cloud: Adv...



A better way of diagnosing the most appropriate treatment is needed. Combining whole genome analysis with artificial intelligence (AI) is a key stepping stone towards fully personalized, highly effective treatment plans.

In recent years, immunotherapy has shown great promise in improving the survival chances of patients with cancer, including those suffering from melanoma or lung cancer.

Immunotherapy works by strengthening a patient's immune system to better defend against cancer cells.

However, a single round of immunotherapy treatment can cost over \$100,000 and is not guaranteed to be effective for all patients.

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—Athon Millane, Machine
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Lead, Max Kelsen

Whole genome sequencing can help predict the effectiveness of cancer treatments, at a fraction of this cost. However, researchers are yet to find an effective way to process whole genome sequences. Part of the processing challenge is the sheer size of each whole genome. At 3.2 billion base-pairs per sequence, each whole genome is almost 300GB. The key to predicting patient treatment outcomes lies in finding and interpreting the patterns and

genes of significance in the genomes of patients who have responded best to previous treatments.

TensorFlow Research Cloud assists with challenge

Through access to [TensorFlow Research Cloud](https://www.tensorflow.org/tfrc/) (<https://www.tensorflow.org/tfrc/>) from Google, which incorporates 1,000 Tensor Processing Units designed for machine learning (ML) as well as AI, [Max Kelsen](https://maxkelsen.com/) (<https://maxkelsen.com/>) is set to address this challenge.

The analytics and software agency aims to use ML and AI to better predict how patients will respond to immunotherapy treatments for cancer.

Google is committed to this leading global research initiative, and the collaboration with Max Kelsen is testament to the quality of research and talent coming out of Australia.

"Access to TensorFlow Research Cloud TPUs will radically transform our capability for this project," says Athon Millane, Machine Learning and Research Lead at Max Kelsen. "Although we have a small team, we are already delivering world-class results. This collaboration with Google provides the ability to explore new techniques and experiments at unprecedented scale."

Analysts predict that by 2025, the volume of genomics data will be 20 times greater than the amount of data on YouTube³. Compute power provided by the TensorFlow Research Cloud helps

researchers leverage this massive influx of data to deliver machine-learning powered advances in a range of fields.

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Immunotherapy Outcome Prediction (IOP)

Immunotherapy Outcome Prediction (IOP) is a project led by Max Kelsen to change the lives of millions of cancer patients worldwide. Working alongside [genomiQa](https://genomiqa.com/) (<https://genomiqa.com/>), [QIMR](https://www.qimrberghofer.edu.au/) [Berghofer](https://www.qimrberghofer.edu.au/) (<https://www.qimrberghofer.edu.au/>), [BGI](#)

(<https://www.bgi.com/us/>), and QLD Health Metro North Hospital and Health Service

(<https://metronorth.health.qld.gov.au/>), Max Kelsen plans to integrate AI and whole genome sequencing into cancer research and clinical practice.

IOP is focusing initially on immunotherapy treatment for melanoma and small cell lung cancer. Max Kelsen aims to leverage the knowledge gained through this project to make a difference to Australians affected by cancer and to take on more healthcare challenges.

Under IOP, BGI will apply DNA sequencing to provide genome data for analysis by genomIQa, a specialist in whole genome sequencing techniques. Max Kelsen will apply AI and ML through the TensorFlow Research Cloud to the analyzed data to better predict patient responses to cancer treatments such as immunotherapy.

The combination of Max Kelsen's expertise in analytics and software engineering with the skills and knowledge of researchers at genomIQa, QIMR, and BGI, and Google compute and ML services, will drive cancer research forward.

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*—Cameron Bean, Health
Lead and
Commercialization Lead,
Max Kelsen*

IOP aims to develop a prediction model that will indicate the likely effectiveness of cancer treatments such as immunotherapy based on each patient's genetic makeup. If successful, the project could help extend patients' lives, increase their quality of life, and eliminate millions of dollars spent on successful treatment costs.

IOP delivers leading approach

"Immunotherapy Outcome Prediction is testament to the leading approach and high level of technical expertise involved in the collaboration between the

organizations involved," says Cameron Bean,
Director of Health Research at Max Kelsen.

Applying ML and AI solutions across industries is
what Max Kelsen specializes in, and personalized
healthcare is next.

"Immunotherapy Outcome Prediction will firmly
establish the role of AI and whole genome analysis
in the future of precision medicine and personalized
healthcare," says Nicholas Therkelsen-Terry, Chief
Executive Officer of Max Kelsen.

¹[Australian Government/Cancer Australia \[sourced from the
Australian Institute of Health and Welfare\]](https://canceraustralia.gov.au/affected-cancer/what-cancer/cancer-australia-statistics/)
([https://canceraustralia.gov.au/affected-cancer/what-
cancer/cancer-australia-statistics/](https://canceraustralia.gov.au/affected-cancer/what-cancer/cancer-australia-statistics/))

²[Bristol-Myers Squibb, Phase 3 Study Evaluating Yervoy®
\(Ipilimumab\) for Melanoma in an Adjuvant Setting Meets Primary
Endpoint of Recurrence-Free Survival](https://news.bms.com/press-release/rd-news/phase-3-study-evaluating-yervoy-ipilimumab-melanoma-adjuvant-setting-meets-pri)
([https://news.bms.com/press-release/rd-news/phase-3-
study-evaluating-yervoy-ipilimumab-melanoma-adjuvant-
setting-meets-pri](https://news.bms.com/press-release/rd-news/phase-3-study-evaluating-yervoy-ipilimumab-melanoma-adjuvant-setting-meets-pri))

³[PLOS Biology, Big Data: Astronomical or Genomical](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4494865/)
(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4494865/>
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